

Drone Fire Control System (FCS) Architecture Specification

Version: 2.0 (Updated Safety & Control Topologies) | **Date:** September 2026

1. System Overview & Physical Infrastructure

This specification outlines the hardware and software architecture for a drone-mounted Fire Control System (FCS). The infrastructure features a ground-based **Panasonic Rugged Tablet / GCS** equipped with a dedicated remote control unit, communicating with an airborne suite driven by a **Jetson Orin Nano** and **two STM32 microcontrollers** connected via a shared CAN Bus loop.

2. Network & Bus Communications

- **Shared CAN Bus Loop:** Due to Jetson Orin Nano hardware constraints for implementing secondary CAN layers, both STM32 units are multi-dropped onto a single, shared CAN Bus channel configured previously. Addressing is isolated via CAN ID ranges: Safety/Firing commands occupy **0x100+**, while Motion/Telemetry occupy **0x200+**. 120Ω termination resistors are placed strictly at the physical terminal ends of the bus.
- **Airborne Ethernet Switch:** To bypass the single Ethernet interface limitation of the GCS wireless data terminal, a miniature, ruggedized airborne Ethernet switch is integrated on the drone. This splits the wireless pipeline between the **Jetson Orin Nano** (receiving tracking vector overrides) and the **Micro300 Optic Master Camera** (streaming low-latency live video).

3. Dual-Switch Safety Activation & Interlocking Logic

The system enforces a strict **Dual-Switch Safety Architecture** mapped directly to physical controls on the remote:

- **Safety 1 (Standby Button):** Activating this hardware control transmits a signal to **STM32 1** to arm the first electronic safety gate. The system enters an armed Standby state but remains incapable of firing.
- **Safety 2 (Trigger Button):** Pressing the trigger closes the second electronic safety gate, completing the weapon discharge loop.
- **Jetson Electronic Interlock:** STM32 1 acts as a hard real-time safety validator. It will refuse to close the safety gates unless the Jetson Orin Nano actively transmits a *'Target Locked & Validated'* status flag over the CAN Bus. Furthermore, if the 2-second Jetson Heartbeat (HB) frame drops, STM32 1 automatically forces both gates open, executing an immediate fail-safe shutdown.

4. System Responsibility & Fail-Safe Matrix

Component	Primary Responsibilities	Hardware & Interfaces	Critical Failsafe Action
GCS Panasonic Tablet & Remote	• Runs target tracking & CV logic (C#) • Transmits target X,Y offsets to Jetson • Relays physical Safety 1 & 2 states	• Ethernet / RF Data-link • Physical Joystick Buttons	Releasing buttons or software crash halts weapon release sequences instantly.

Component	Primary Responsibilities	Hardware & Interfaces	Critical Failsafe Action
■ FCS Jetson Orin Nano	- Acts as Mission & Ballistic computer - Integrates X,Y frames with LRF data - Computes precise gimbal trajectories - Emits electronic target validation lock	- Airborne Ethernet Switch - Laser Range Finder (LRF) - Shared CAN Bus line	If GCS data-link drops, drops target validation and ceases Heartbeat (HB) frames immediately.
■ STM32 1 (FCS/ISM 1)	- Manages physical safety gates - Decodes remote Safety 1 & 2 states - Validates clicks against Jetson interlocks - Monitors weapon recoil behavior	- Firing Solenoid / Actuator - Recoil Sensor - Shared CAN Bus line	If Jetson HB drops (>2s) or recoil anomaly is detected, locks all safety gates natively.
■ STM32 2 (FCS/ISM 2)	- Drives low-level motion control (PID loops) - Executes Azimuth & Elevation movement - Aggregates environmental telemetry	- AZ/EL Servos & IMU - Angle Sensors - Temp Sensor	Natively freezes and kills servo power if internal temperature thresholds cross 100°C.